Intestinal Flukes- *Heterophyes*

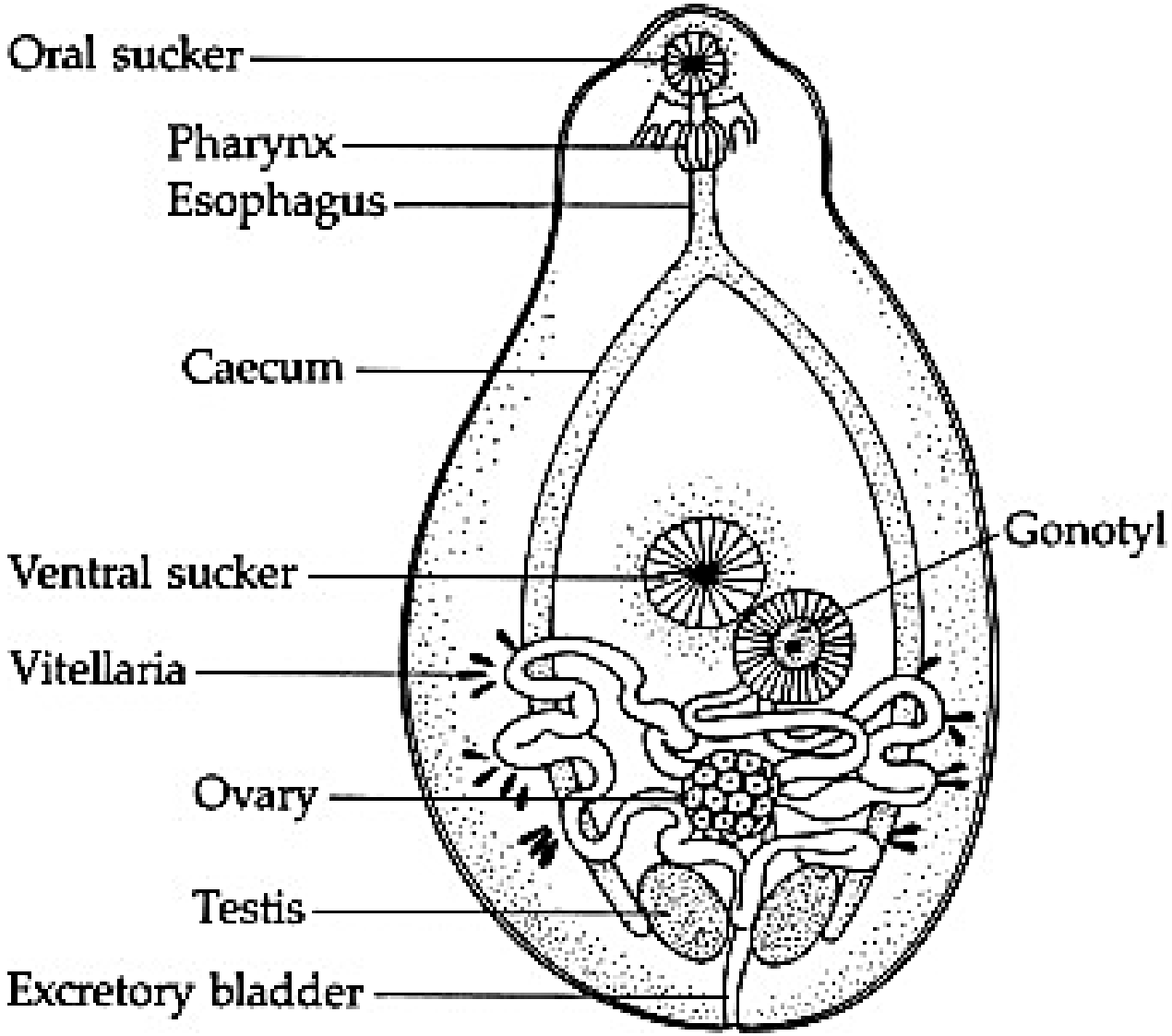
Heterophyes heterophyes

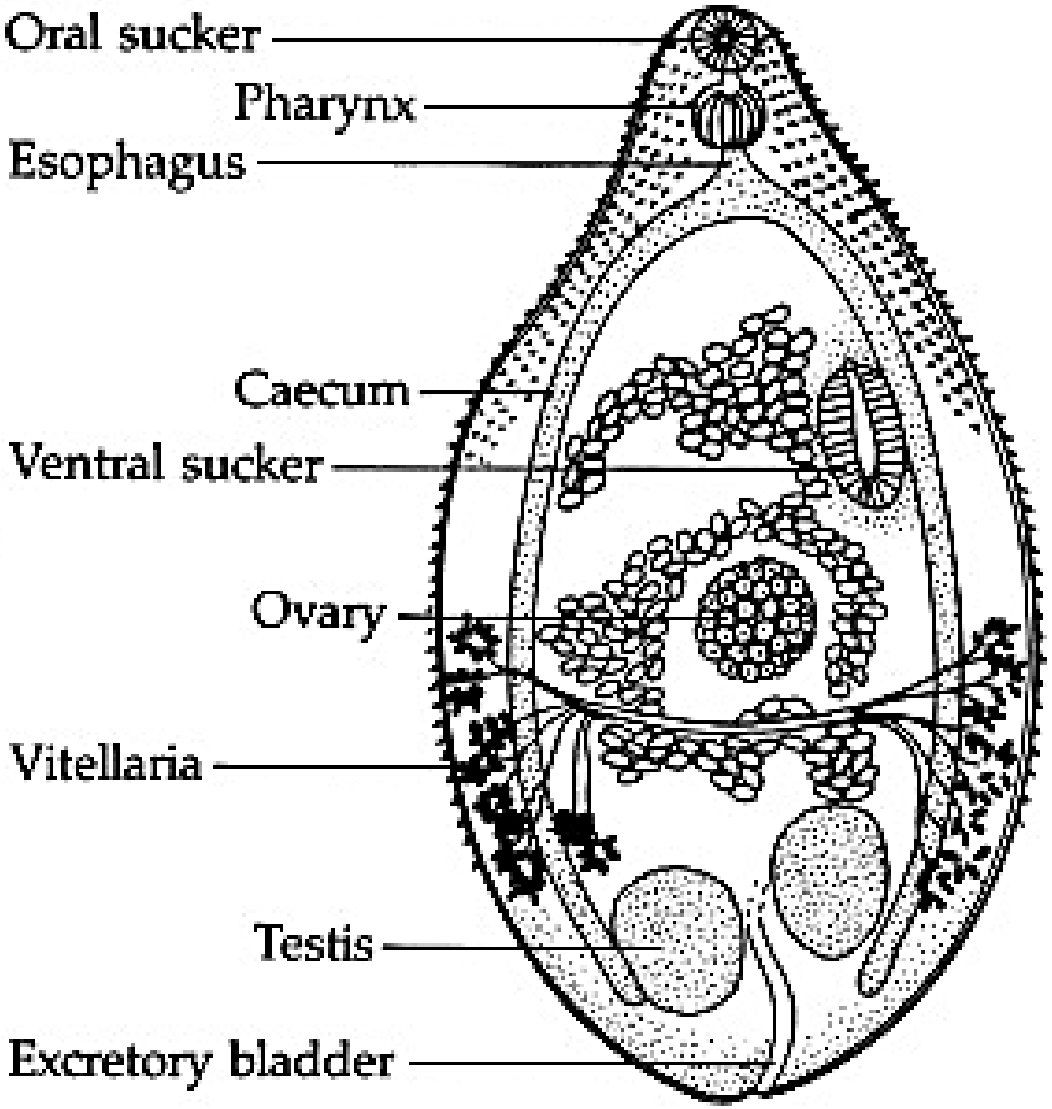
- **Location of adult:** Small intestine.
- **Intermediate host:**

  Primary: *Pirenella conica.*

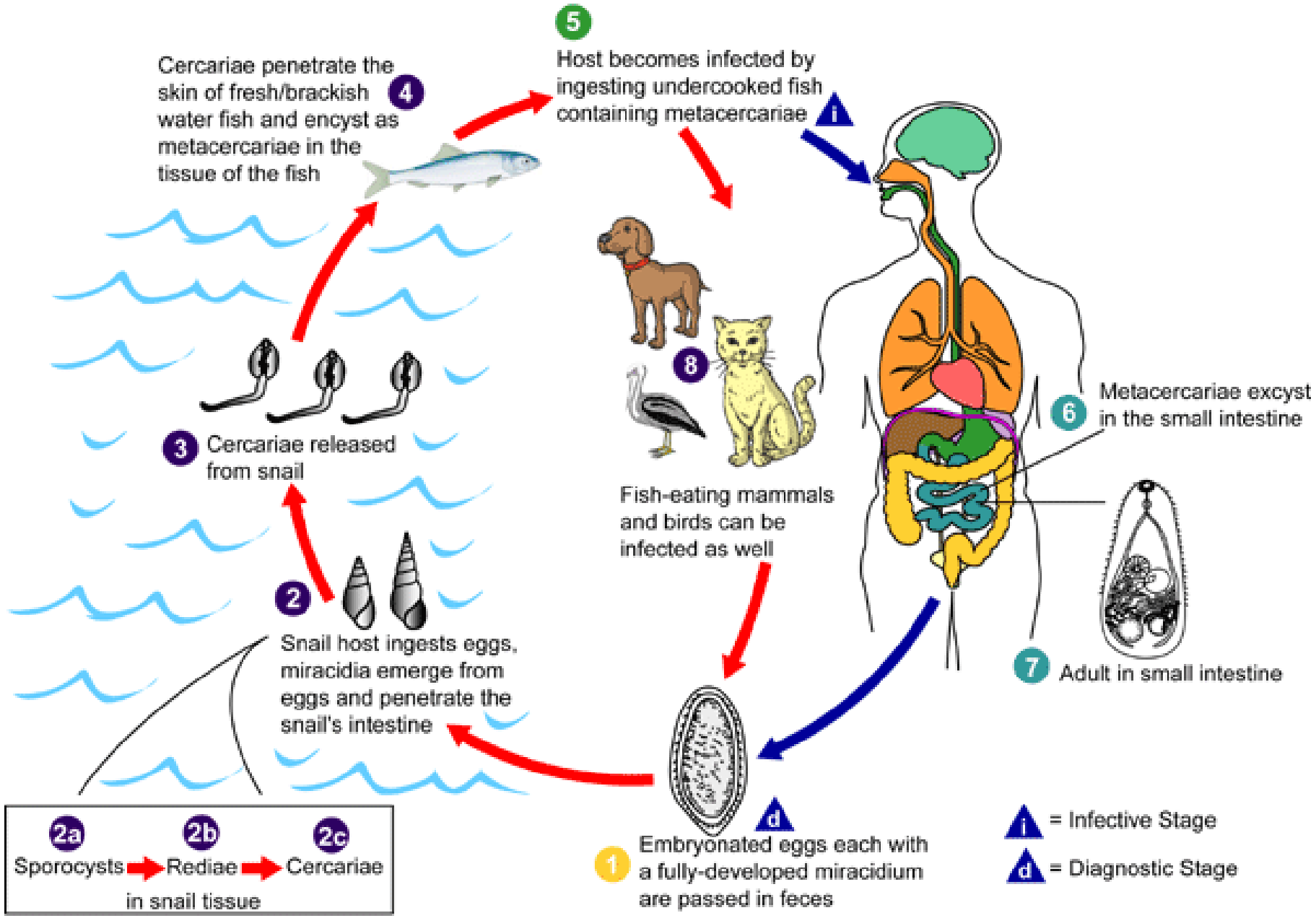
  Secondary: Bolti and Bouri fish.
- **Infective stage**: Encysted metacercaria.
- **Mode of transmission**: Ingestion of raw or undercooked fish containing encysted metacercaria.
- **Diagnosis:** Eggs in stool.
- **Disease:** Heterophiasis.

Heterophyes-Reproductive system





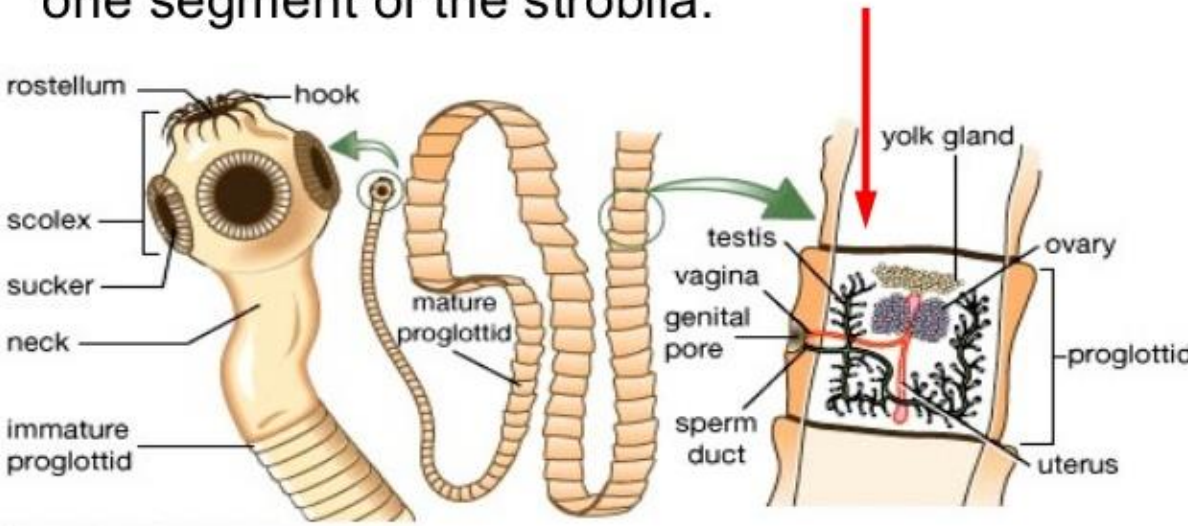
Heterophyes-Life Cycle



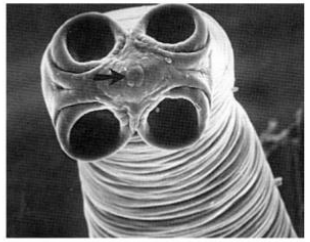
Class Cestoda

- consist of a rounded head, called a scolex, and long strobila or chain of proglottids (multiple segments) of varying stages of maturity.
- They have no digestive tract of its own at any point in its life cycle.
- The scolex has specialized means of attaching to the intestinal wall, namely suckers, hooks, or sucking grooves.
- All species are parasitic during both the adult and larval developmental stages.
- All cestodes have stage of oncosphere in the life cycle.

- Contains both male and female organs.
- Essentially a whole reproductive package in one segment of the strobila.

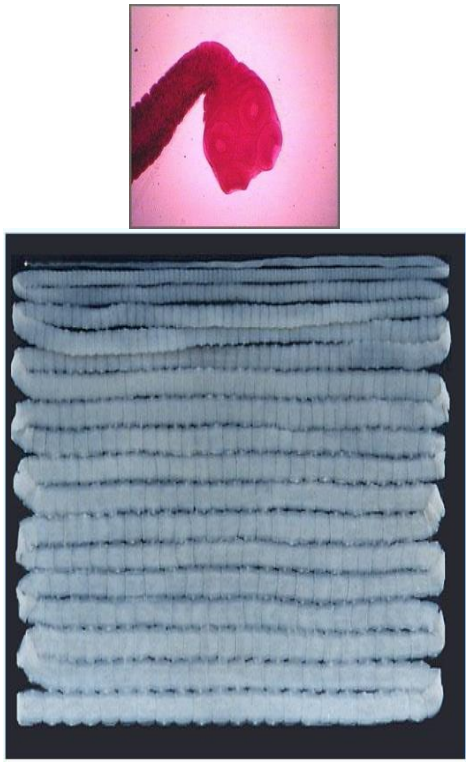


Suckers, Hooks and Bothria (with slit-like groove with weak suction powers and usually two in number)

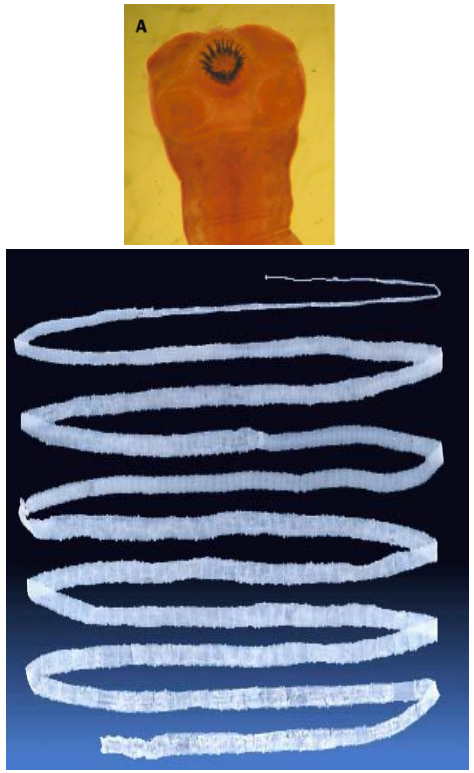


Taenia

- ***Taenia saginata***, which is also known as the "beef tapeworm", though it also infects humans.
- ***Taenia solium***, which is known as the "pork tapeworm". Like *Taenia saginata* humans serve as its primary host.
- Both have mature, immature and gravid segments.
- Both have scolex provided with four cup- shaped suckers, but *T. saginata* has an unarmed scolex while *T. solium* has an armed scolex.
- In both species the infective stage is the cysticercus.
- Cows are the intermediate host of *T. saginata* while pigs are the intermediate host of *T. solium.*
- Definitive host: Human.
- Diagnosis:
  - By finding gravid proglotid or eggs in stool.
    - Eggs present in stool less often are proglottids.
  - Direct fecal smear.



Taenia saginata

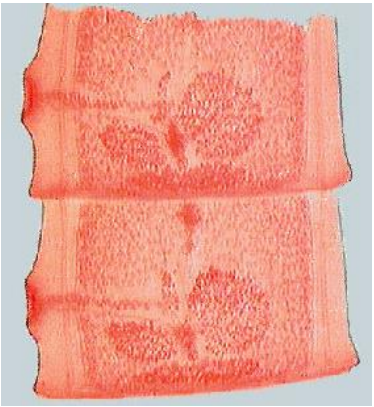


Taenia solium

Taenia-Morphology

Aspect	Taenia saginata	Taenia solium
Intermediate host	Cows (Beef tape worm)	Pigs (Pork tape worm)
Adult size	Longer (4 – 8 meters)	Smaller (2 - 4 meters)
Number of segment	1000 – 2000	700 - 1000
Mature segment	2 lobes ovary	3 lobes ovary
Gravid proglotid	With 15 – 30 uterine branches	With 7 – 13 uterine branches
Scolex	Devoid rostellum and hooks	With rostellum and 2 rows of hooks
Larvae	Cysticercus bovis in cattle only	Cysticercus cellulosae in pig as well as in man
Disease caused in man	Taeniasis	Taeniasis and cysticercosis

Taenia saginata
2 ovary lobes



Taenia solium
3 ovary lobes

Taenia-Reproductive system

➢ Male reproductive system

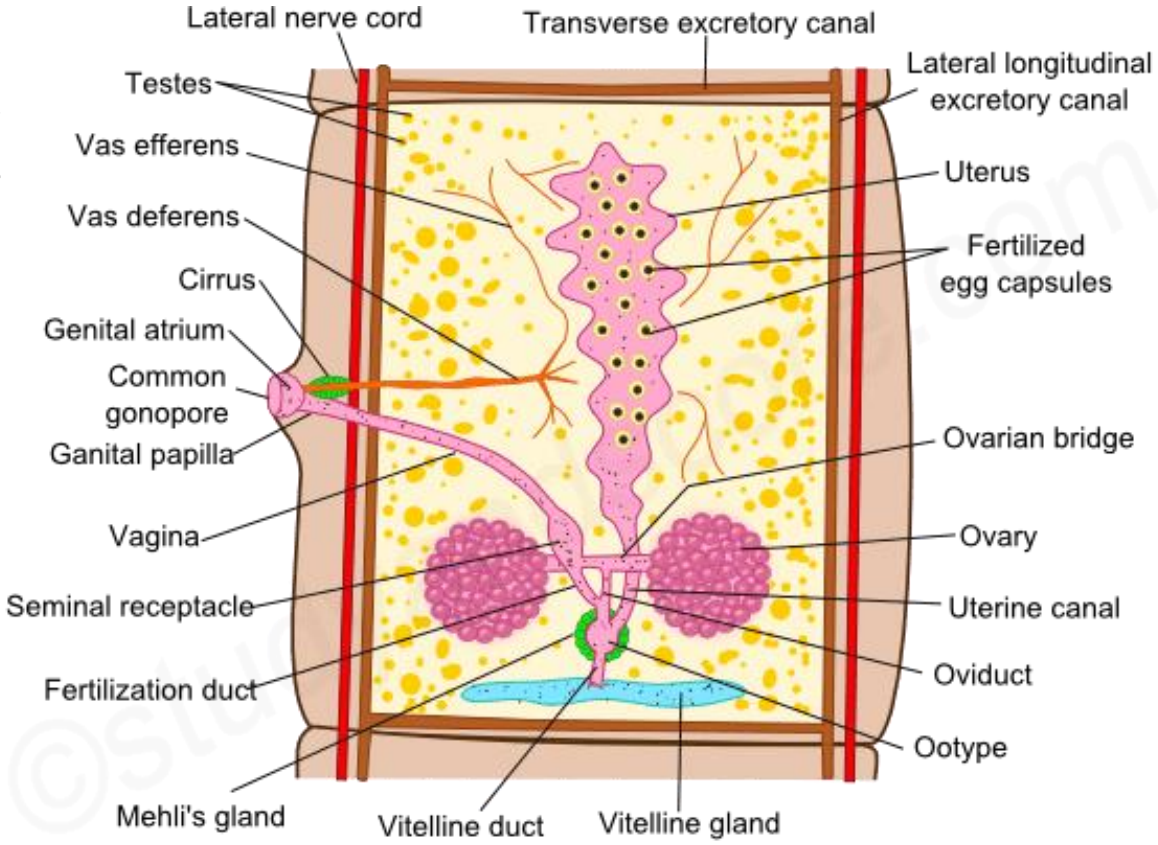
Testes: They are numerous, small, spherical bodies scattered throughout the mesenchyme close to the dorsal surface.

Vas efferentia: From each testis a find ductule called as vas efferentia arises. Vas efferentia usually gets interconnected with similar ductules from the surrounding of the testes.

Vas deferens: It is a thick and convoluted tube extending up to the lateral margin of proglottid at right angles to it.

Cirrus: Outer end of vas deferens forms lumen of a thick, muscular copulatory organ called as the cirrus.The seminal vesicles are absent in tapeworm.

Genital atrium: The cirrus opens into a cup-shaped genital atrium through male genial pore. Genital atrium in turn opens to outside through a common gonopore. The genital papilla in the middle of the lateral margin of proglottid. Gonopore is provided with the sphincter to control the flow of the gonads. The common gonopores are arranged alternatively on both sides of the proglottid.



TAENIA SOLIUM - REPRODUCTIVE SYSTEM IN MATURE PROGLOTTID

Taenia-Reproductive system

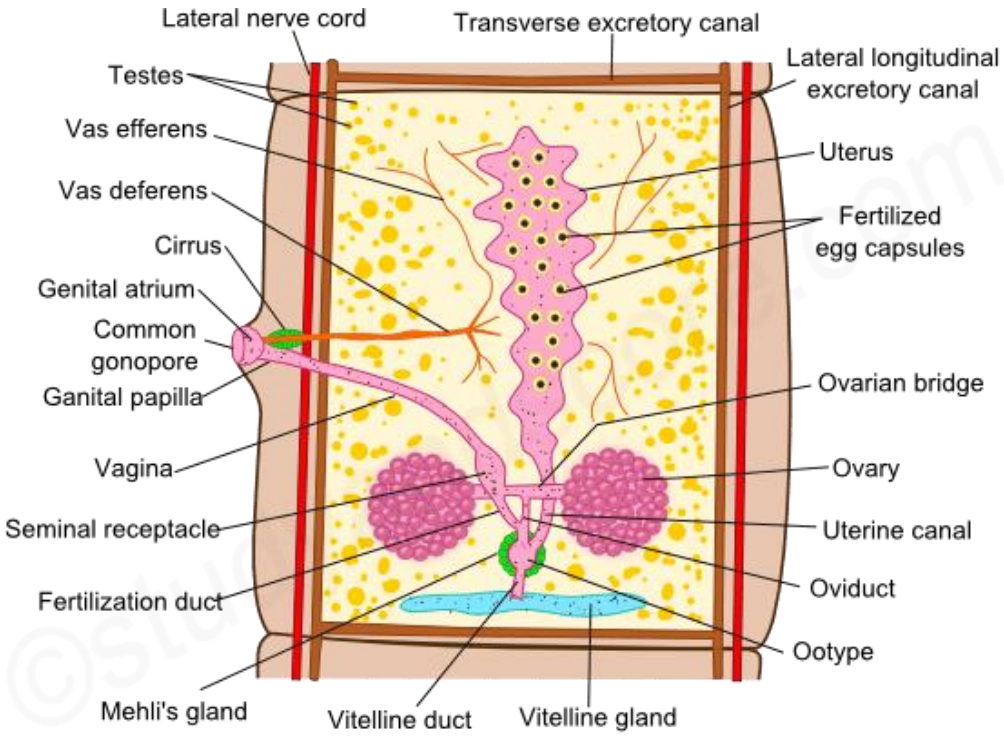
➢ Female reproductive system

Ovary: It is also called germarium. Ovary is a bilobed organ situated ventrally in the posterior part of proglottid. Both the lobes are connected by a transverse ovarian bridge.

Oviduct: It arises from the middle of ovarian bridge and opens into the ootype.

Ootype: It is spherical bulb-like structure situated at the junction of oviduct, uterus and vitelline duct. The seminal receptacle stores the sperms temporarily. The unicellular Mehli's glands surround the ootype and secrete a slimy substance that lubricates the eggs in uterus.

Vagina: It is absent narrow tube. It originates from the female genital pore located behind the male genital pore in genital atrium. Vagina swells to form a sperm-storing sac.

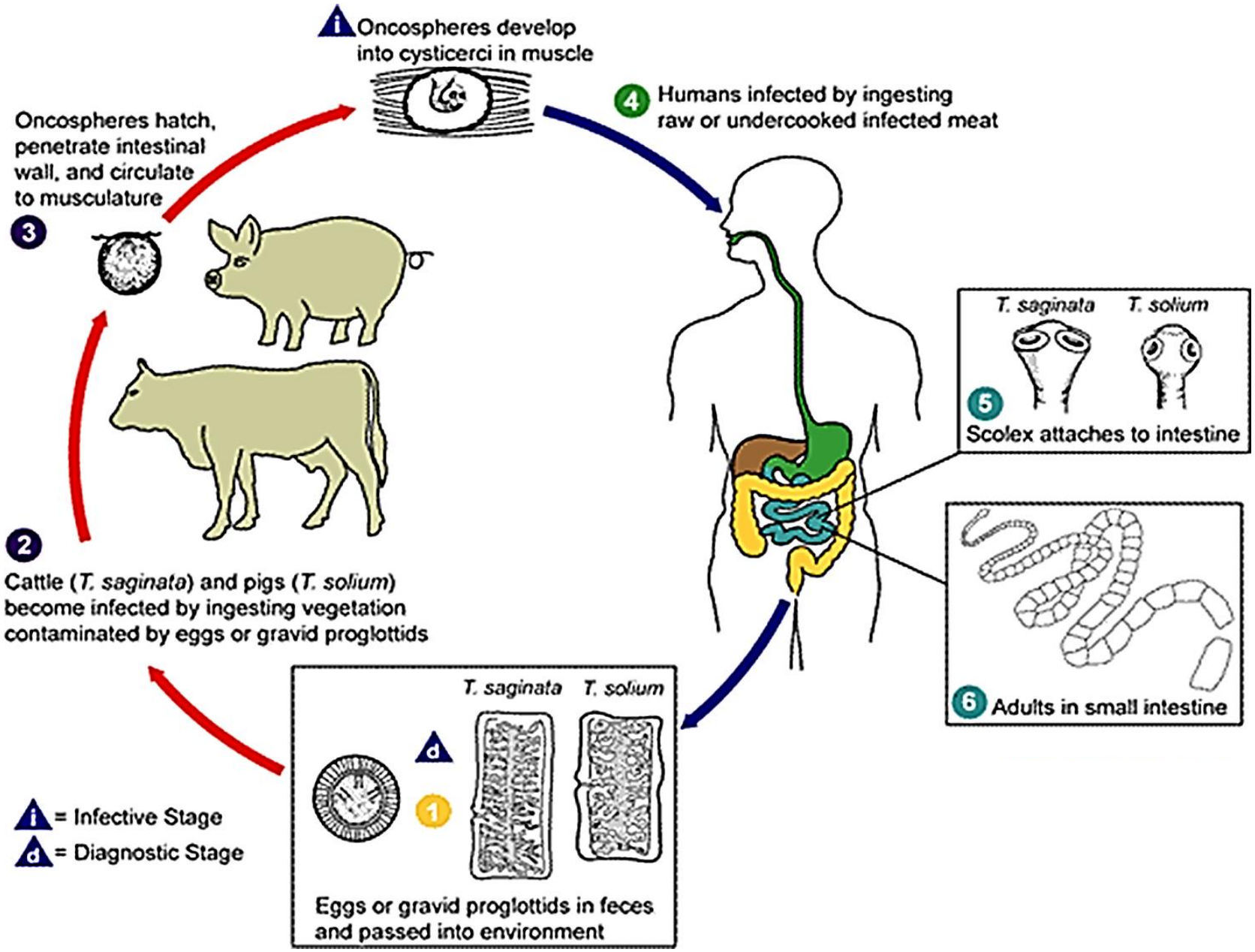


TAENIA SOLIUM - REPRODUCTIVE SYSTEM IN MATURE PROGLOTTID

Uterus: It is a blind sac-like structure originating from the ootype and extending towards the anterior side of the proglottid. In gravid proglottids uterus becomes large and highly branched with eggs in various developmental stages.

Vitelline gland: It is situated on the posterior margin of the proglottid and connected to the ootype. It connects with the ootype by a short vitelline duct. It consists of numerous follicles secreting yolk cells.

Taenia-Life cycle



Nematoda

- Has a two-hole digestive tract.
- They lack body segmentation.
- Colorless
- Range in length from microscopic to several meters long.
- May be free-living or parasitic.

- Nematodes have three cell layers:

  Ectoderm – outer tissue layer (epidermis)

  Mesoderm – middle tissue layer (muscle)

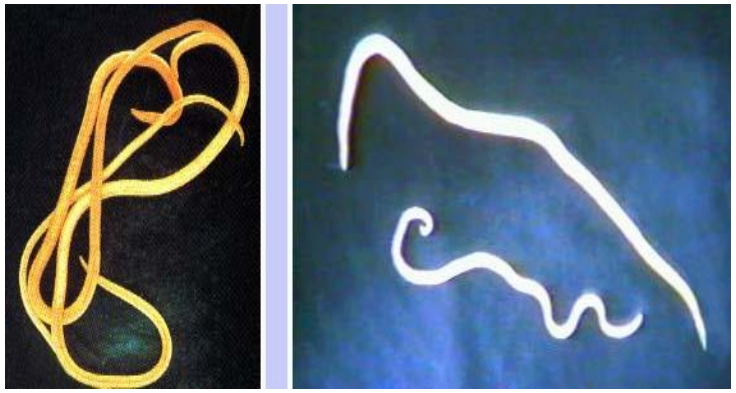
  Endoderm – innermost tissue layer (body cavity)

- They are known as psuedocoelomates because they have an internal cavity that is not lined with peritoneum – therefore it is not a “true” coelom.

- They are bilaterally symmetrical – as is every organism we study from here on out.

- “thread-like”, commonly called roundworms, found just about everywhere

- Examples: ascaris, pinworms, hookworms, trichina worms



Adult worm of *Ascaris lumbricoides*

Ascaris lumbricoides

- *Ascaris lumbricoides* is the largest nematode (roundworm) parasitizing the human intestine.
- Ascaris lumbricoides is an intestinal worm found in the small intestine of man.
- They are more common in children then in adult.
- As many as 500 to 5000 adult worms may inhabit a single host.

Common name:- Giant Intestinal round worms.

Disease:-Ascariaisis

Host :- The human is intermediate and final host.

Location in Definitive host :-

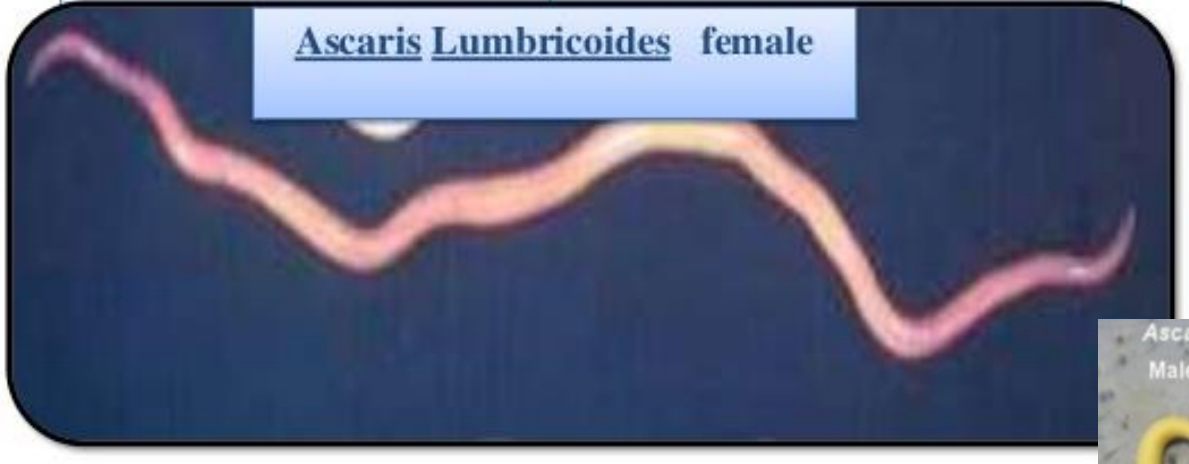
- **the adult worm: in small intestines .**
- **larva: in lung .**

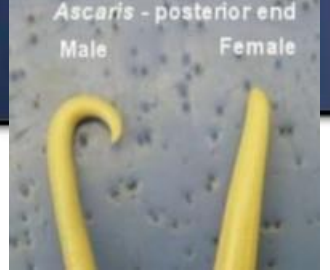
Infective stage :- Ovum

Ascaris lumbricoides

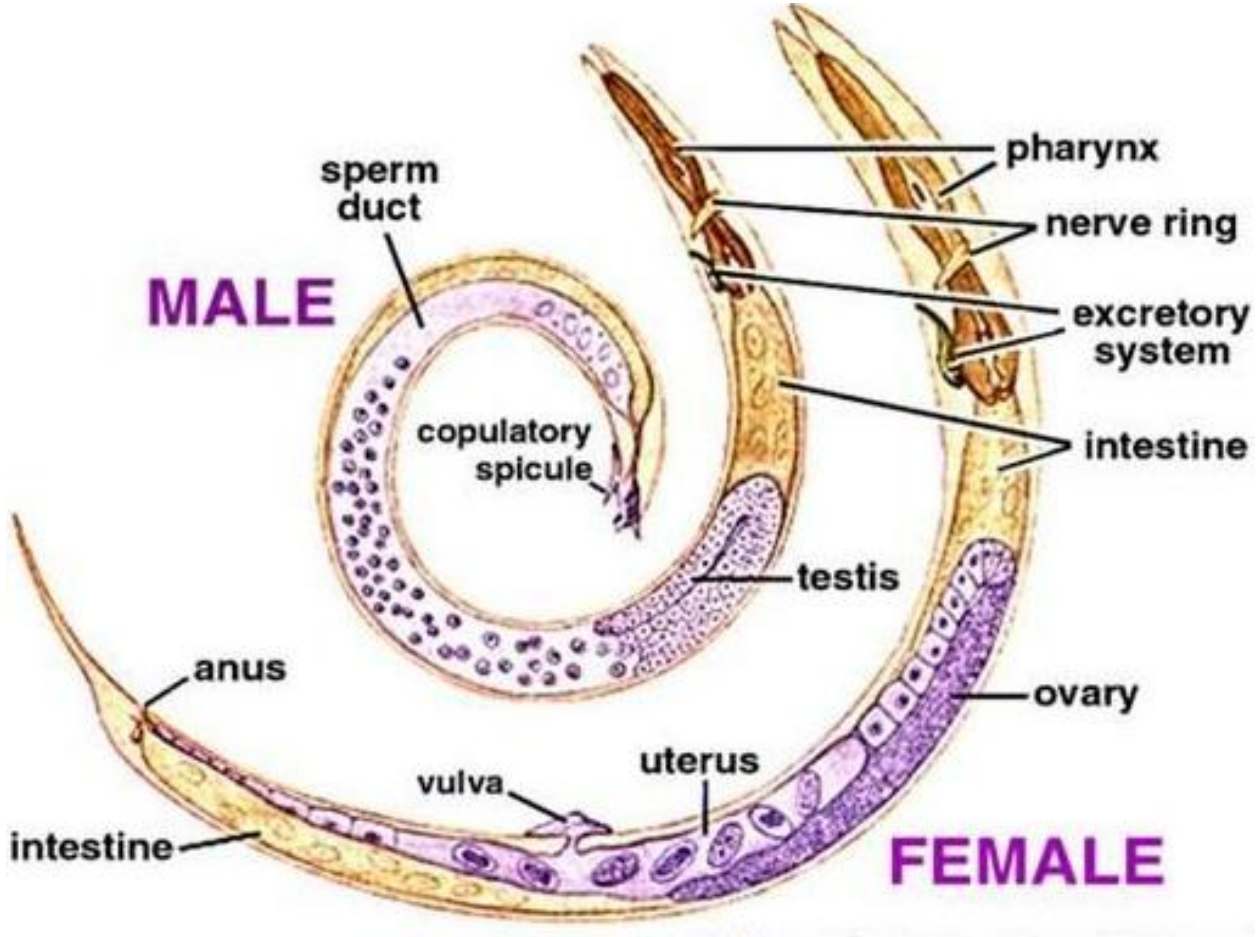
Characteristics	Female
1.Size	200 –350 mm X 4- 6 mm
2.Color	Ivory mixed light red
3.Posterior end	Straight
4. Spicules	Absent
5.Papillae	Absent

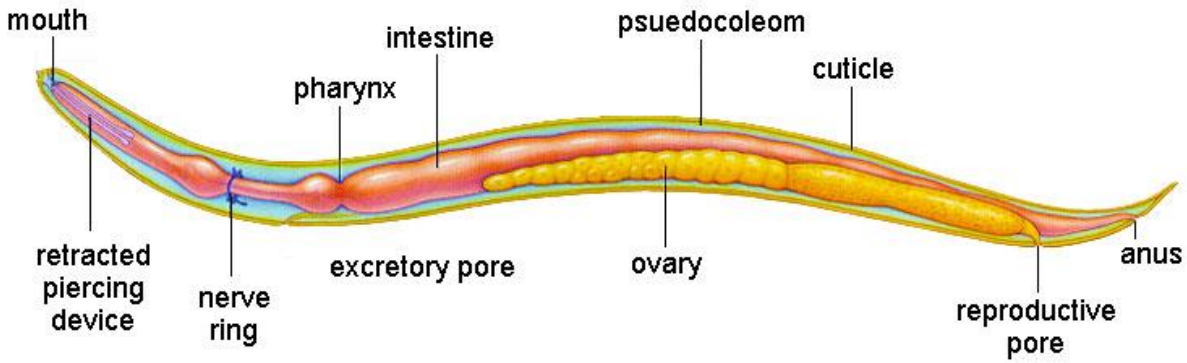
Characteristics	Male
1.Size	150 – 200 X 2 -4 mm
2.Color	Reddish pink
3.Posterior end	Curved
4. Spicules	Has two spicules
5.Papillae	Has many from papillae





Ascaris lumbricoides



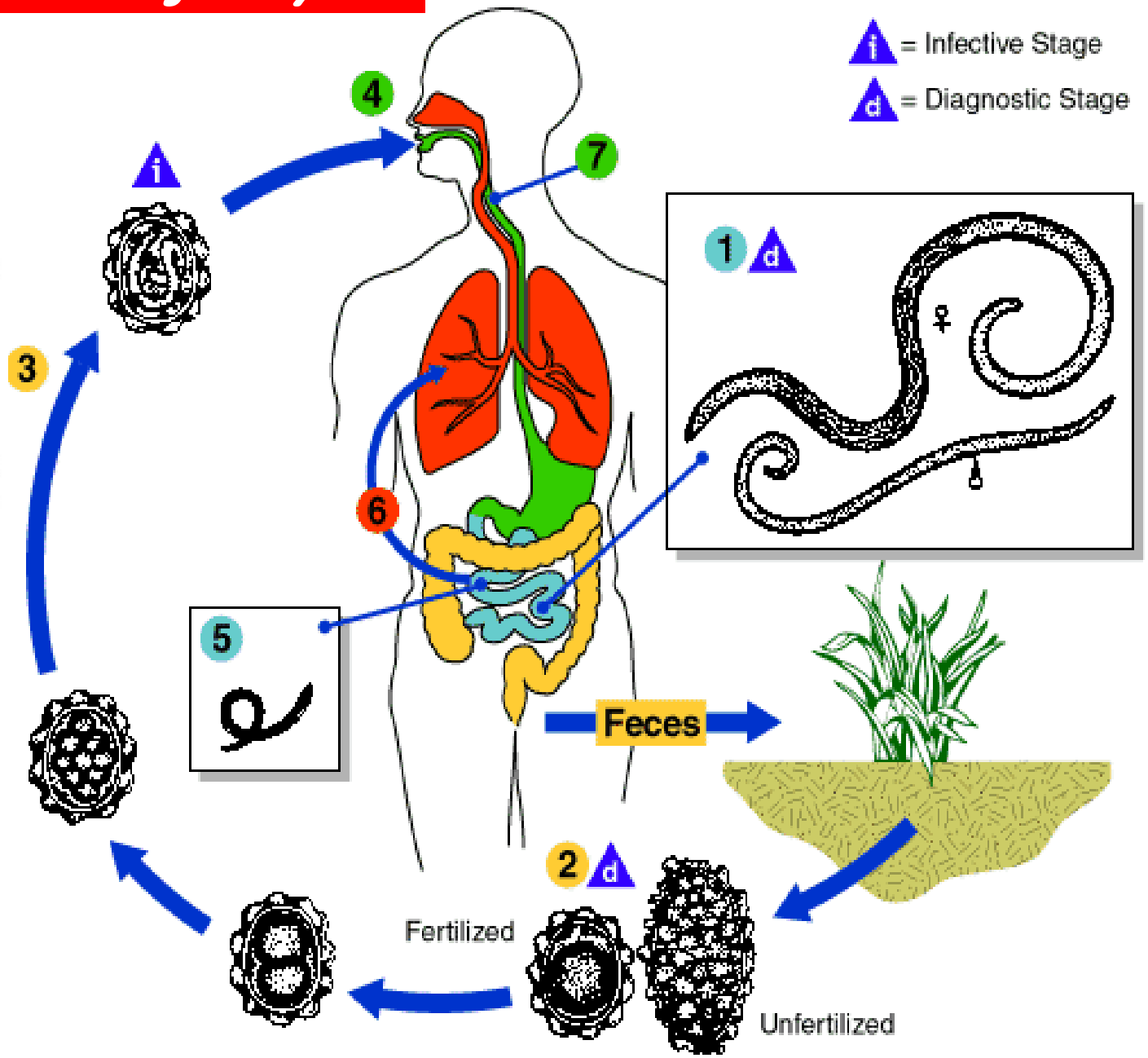


Ascaris-life cycle

- Adult worms *(1)* live in the lumen of the small intestine. A female may produce approximately 200,000 eggs per day, which are passed with the feces.

- *(2).* Unfertilized eggs may be ingested but are not infective. Fertile eggs embryonate and become infective after 18 days to several weeks **(3),** depending on the environmental conditions (optimum: moist, warm, shaded soil).

- After infective eggs are swallowed *(4),* the larvae hatch **(5),** invade the intestinal mucosa, and are carried via the portal, then systemic circulation to the lungs

- *(6)* The larvae mature further in the lungs (10 to 14 days), penetrate the alveolar walls, ascend the bronchial tree to the throat, and are swallowed (7) Upon reaching the small intestine.

- They develop into adult worms; Between 2 and 3 months. Adult worms can live 1 to 2 years.